8.4 Trigonometric Substitutions

- For integrals involving
  $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ and $\sqrt{x^2 - a^2}$, where $a > 0$.

- Three basic trigonometric substitutions are
  $x = a\sin\theta$, $x = a\tan\theta$ and $x = a\sec\theta$.

When a substitution is done, we need it to be reversible, and by recalling inverses of trigonometric functions, a substitution can be done under certain domain constraints.

- Using the substitution of $x = a\sin\theta$ in $\sqrt{a^2 - x^2}$, we require $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

- Using the substitution of $x = a\tan\theta$ in $\sqrt{a^2 + x^2}$, we require $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

- Using the substitution of $x = a\sec\theta$ in $\sqrt{x^2 - a^2}$, we require $0 \leq \theta < \frac{\pi}{2}$ if $\frac{x}{a} \geq 1$ and $\frac{\pi}{2} < \theta \leq \pi$ if $\frac{x}{a} \leq -1$.

Remark:

- For $\sqrt{a^2 - (bx)^2}$, substitute $x = \frac{a}{b}\sin\theta$.
- For $\sqrt{a^2 + (bx)^2}$, substitute $x = \frac{a}{b}\tan\theta$.
- For $\sqrt{(bx)^2 - a^2}$, substitute $x = \frac{a}{b}\sec\theta$.

Example

$$\int \frac{dx}{\sqrt{4+x^2}} =$$

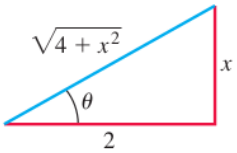


FIGURE 8.4 Reference triangle for $x = 2\tan\theta$ (Example 1):

$$\tan\theta = \frac{x}{2}$$

and

$$\sec\theta = \frac{\sqrt{4+x^2}}{2}.$$

Example

$$\int \frac{x^2 dx}{\sqrt{9-x^2}} =$$

Example

$$\int \frac{dx}{\sqrt{25x^2-4}} =$$

Example

Find the volume of the solid generated by revolving about the x-axis the region bounded by the curve $y = 4/(x^2 + 4)$, the x-axis, and the lines $x = 0$ and $x = 2$.

Example

Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Extra Examples:

1. $\int \sqrt{9 - 4x^2}\, dx = \frac{1}{2}x\sqrt{9 - 4x^2} + \frac{9}{4}\sin^{-1}\frac{2x}{3} + C$

2. $\int_0^{\sqrt{3}/2} \frac{4x^2}{(1-x^2)^{3/2}}\, dx = 4\sqrt{3} - \frac{4\pi}{3}$

3. $\int \frac{2}{\sqrt{t}+4t\sqrt{t}}\, dt = 2\tan^{-1}(2\sqrt{t}) + C$